Quantum and Semiclassical Molecular Dynamics

Computational Exercises

Dr. David Picconi

Heinrich-Heine University Düsseldorf

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Introduction

This document contains computational exercises involving the simulation of different types of photoinduced nonadiabatic processes and their related spectroscopic observables. The simulations will be performed using either **quantum dynamical** or **semiclassical** methods.

Each exercise includes step-by-step **tutorials** to compute specific observables using dedicated software, as well as **open-ended questions** and pen-and-paper exercises. At the end of each exercise, guidance is briefly provided on how the corresponding **protocol** should be prepared. In particular the answers to the questions should be present in the protocol.

1.1 General nomenclature

Quantum dynamical simulations involve the solution of the time-dependent Schrödinger equation

$$\frac{\partial|\Psi,t\rangle}{\partial t} = -\frac{\mathrm{i}}{\hbar}\hat{H}|\Psi,t\rangle \,, \tag{1.1}$$

where the **molecular wavefunction** is expanded in a basis of diabatic electronic states,

$$|\Psi,t\rangle = \sum_{\alpha=1}^{N_{\mathrm{el}}} |\alpha\rangle\chi_\alpha(\mathbf{q},t) \,, \tag{1.2}$$

where $\chi_\alpha(\mathbf{q},t)$ are time-dependent nuclear wavefunctions (or **"wave packets"**), and $\mathbf{q} = (q_1,...,q_f)$ is a vector containing the nuclear coordinates.

In the diabatic representation the **molecular Hamiltonian** $\hat{H}$ is a sum of the kinetic energy operator $\hat{T}$ and electronic operators $|\alpha\rangle\langle\beta|$ each associated with a potential function $W_{\alpha\beta}(\mathbf{q})$ dependent on the nuclear coordinates,

$$\hat{H} = \hat{T} + \sum_{\alpha,\beta=1}^{N_{\mathrm{el}}} |\alpha\rangle\langle\beta|W_{\alpha\beta}(\mathbf{q}) \,. \tag{1.3}$$

For ease of nomenclature, the "diagonal" terms $W_{\alpha\alpha}(\mathbf{q})$ are denoted *diabatic state potentials*, and the "off-diagonal" terms $W_{\alpha\beta}(\mathbf{q})$ with $\alpha \neq \beta$ are called *diabatic couplings*. In all tutorials rectilinear coordinates are used, so that the kinetic energy operator has the form

$$\hat{T} = \sum_{\kappa=1}^{f} -\frac{\hbar^2}{2m_\kappa}\frac{\partial^2}{\partial q_\kappa^2} \,, \tag{1.4}$$

where the "masses" m_κ depend on the specific type of coordinate system used in each exercise. In particular, some tutorials use dimensionless coordinates, in which case one writes $m_\kappa = \hbar/\omega_\kappa$, where ω_κ is the frequency associated to the κ-th coordinate.[1]

<u>**Semiclassical simulations**</u> are performed in the **adiabatic** representation. Here the wavefunction is purely electronic,

$$|\Psi, t\rangle = \sum_{\alpha=1}^{N_{\text{el}}} C_\alpha(t) \, |\alpha\left(\mathbf{q}(t)\right)\rangle \ , \tag{1.5}$$

and $|\alpha\left(\mathbf{q}(t)\right)\rangle$ denotes an **adiabatic** electronic state, i.e. a state formally defined as an eigenstate of the electronic Hamiltonian. Each adiabatic state is associated to a potential energy surface $V_\alpha(\mathbf{q})$ and depends on the nuclear coordinates $\mathbf{q}(t) = \left(q_1(t), ..., q_f(t)\right)$ that evolve in time according to **Newtonian trajectories**.

The complex coefficients $C_\alpha(t)$ follow coupled time-dependent Schrödinger equations in the form

$$\frac{\mathrm{d}C_\alpha(t)}{\mathrm{d}t} = -\frac{\mathrm{i}}{\hbar} V_\alpha\left(\mathbf{q}(t)\right) C_\alpha(t) - \sum_{\beta\neq\alpha} \sum_{\kappa=1}^{f} D_\kappa^{(\alpha\beta)}\left(\mathbf{q}(t)\right) \frac{\mathrm{d}q_\kappa(t)}{\mathrm{d}t} C_\beta(t) \ , \tag{1.6}$$

where

$$D_\kappa^{(\alpha\beta)}(\mathbf{q}) = \left\langle \alpha(\mathbf{q}) \left| \frac{\partial}{\partial q_\kappa} \right| \beta(\mathbf{q}) \right\rangle \tag{1.7}$$

are the **derivative couplings** between the adiabatic states $|\alpha(\mathbf{q})\rangle$ and $|\beta(\mathbf{q})\rangle$.

The nuclear coordinates for various trajectories need to be sampled from some initial distribution (thermal, Wigner, etc.) and are propagated in time according to Newton's equations that have the general form

$$\frac{\mathrm{d}^2 q_\kappa(t)}{\mathrm{d}t^2} = -\frac{1}{m_\kappa} \frac{\partial V_{\text{eff}}(\mathbf{q})}{\partial q_\kappa} \ , \tag{1.8}$$

where the potential $V_{\text{eff}}(\mathbf{q})$ depends on the specific semiclassical method. In **surface hopping** approaches V_{eff} coincides, at each time, with the adiabatic potential V_α of the so called "active state" and a stochastic (or, in special methods, deterministic) algorithm is use decide when the active state should change and how the velocity should be rescale to ensure energy conservation. In the **Ehrenfest** approach V_{eff} is given as the population-weighted average $V_{\text{eff}} = \sum_{\alpha=1}^{N_{\text{el}}} |C_\alpha(t)|^2 \, V_\alpha(\mathbf{q})$.

Similar to the case of quantum dynamics, the "mass" m_κ appearing in Eq. (1.8) depends on special coordinate system used. If dimensionless coordinates are used in semiclassical dynamics, the parameter m_κ becomes $\hbar/\omega_\kappa$.

[1] Note that, in this case, the parameter m_κ does not even have the units of a mass.

1.2 Software

The exercises of this computational laboratory are designed to be performed using the Heidelberg `MCTDH` program or `Quantics` for quantum dynamics, and scripts relying on the package `PySurf` for semiclassical dynamics. The program `GNUPlot` is also used sometimes for plotting purposes.

The tutorials are illustrated for a standard Linux shell and specific indications on shell commands are highlighted inside a yellow box. For example, the following box illustrates the usage of some common linux commands.

```
# Create a folder named 'test'
$ mkdir test

# Move into the folder
$ cd test

# Move one level up
$ cd ..

# Copy a folder and all its contents
$ cp -r test test_copy

# Rename a folder (or a file)
$ mv test test_renamed

# List the contents of the current folder
$ ls
test_copy test_renamed

# Remove a folder (or a file) and all its contents
# WARNING: this operation is irreversible (no trash bin)
$ rm -r test_copy
```

Tutorial 1. Photoelectron spectrum of butatriene

The ionisation of butatriene in the lowest-energy photoelectron band creates the butatriene cation in a molecular state that spans two electronic states. In the diabatic representation the elecronic states have symmetry B_{2g} and B_{2u}.

The goal of this exercise is twofold:

1. To simulate the time-evolution of the molecular state, comparing results obtained by two trajectory-based methods – Tully's surface hopping and Ehrenfest dynamics – with reference observables obtained by full quantum dynamics.

2. To simulate the photoelectron spectrum using the time-dependent formalism for spectroscopy.

To this end, we use the quadratic vibronic coupling model constructed by Cattarius *et al.*[1] where the diabatic potentials are obtained by a second order expansion around a reference geometry ($q_\kappa = 0$),

$$W_{\alpha\beta}(\mathbf{q}) = W^{(0)}_{\alpha\beta} + \sum_{\kappa=1}^{f} W^{(\kappa)}_{\alpha\beta} q_\kappa + \frac{1}{2} \sum_{\kappa,\kappa'=1}^{f} W^{(\kappa\kappa')}_{(\alpha\beta)} q_\kappa q_{\kappa'} \quad \text{for } \alpha, \beta = B_{3g}, B_{2g} \,. \tag{2.1}$$

Here, the coordinates q_κ are *dimensionless normal modes*. Using these coordinates the kinetic energy operator takes the form

$$\hat{T} = -\sum_{\kappa=1}^{f} \frac{\hbar\omega_\kappa}{2} \frac{\partial^2}{\partial q_\kappa^2} \,. \tag{2.2}$$

The initial vibrational state in the Franck-Condon approximation is taken as the vibrational ground state of the neutral butatriene. Using dimensionless normal modes this state has the simple form

$$\chi_0(\mathbf{q}) = \frac{1}{\pi^{\frac{f}{4}}} \prod_{\kappa=1}^{f} \mathrm{e}^{-\frac{q_\kappa^2}{2}} \,. \tag{2.3}$$

A. Answer the following questions.

[1] Chr. Cattarius, G. A. Worth, H.-D. Meyer and L. S. Cederbaum, *J. Chem. Phys.* **115**, 2088 (2001)

Question 1. What is the point symmetry group of butatriene? How many degrees of freedom (f) does the molecule have?

Question 2. The diabatic potentials are formally given as $W_{\alpha\beta}(\mathbf{q}) = \langle \alpha | \hat{H}_{\mathrm{el}}(\mathbf{q}) | \beta \rangle$, where $\hat{H}_{\mathrm{el}}(\mathbf{q})$ is the electronic Hamiltonian (totally symmetric) and $|\alpha\rangle$ and $|\beta\rangle$ denote diabatic states which can have symmetry $B_{3\mathrm{g}}$ or $B_{2\mathrm{g}}$. What is the symmetry (i.e. the irreducible representation) of the diagonal diabatic potentials $W_{\alpha\alpha}(\mathbf{q})$ and the off-diagonal one $W_{\alpha\beta}(\mathbf{q})$ (with $\alpha \neq \beta$)?

B. We start from the quantum dynamical simulation of the butatriene cation following the ionisation to the $B_{2\mathrm{u}}$ state. This is performed by solving the time-dependent Schrödinger equations for the coupled electronic-vibrational evolution using the `MCTDH` package. In the MCTDH approach the vibrational wavefunction $\chi_\alpha(\mathbf{q}, t)$ for each electronic state $|\alpha\rangle$ is given as a sum of Hartree products,

$$\chi_\alpha(\mathbf{q}, t) = \sum_{j_1=1}^{n_1^{(\alpha)}} \sum_{j_2=1}^{n_2^{(\alpha)}} \cdots A^{(\alpha)}_{j_1 j_2 \ldots}(t) \varphi^{(1,\alpha)}_{j_1}(\mathbf{Q}_1, t) \varphi^{(2,\alpha)}_{j_2}(\mathbf{Q}_2, t) \cdots , \tag{2.4}$$

where each "combined mode" $\mathbf{Q}_r$ accounts for a subset of M "physical modes" and the single-particle functions (SPFs) $\varphi^{(r,\alpha)}_{j_r}(\mathbf{Q}_r)$ are represented on a $M-$dimensional grid.

The details of the wavefunctions are specified in the input file for the `MCTDH` program. To see it, move into the folder `exercise_1/quantum_dynamics`; the folder contains the input file `C4H4_B2u.inp` and the operator file `C4H4.op`. Open the input file with a text editor and, by inspecting it, answer the following questions.

Question 3. The wavefunction is specified, for each of the two diabatic states, in the part starting from `SPF_BASIS_SECTION`. (a) How many combined modes are used? Which physical modes are accounted for by each of them? (b) What is the number of SPFs $n_r^{(\alpha)}$ that is used for each combined mode $\mathbf{Q}_r$ and each electronic state? (c) How many configurations compose the vibrational wavefunction for each diabatic state?

Question 4. What is the total propagation time requested in the input? What is the "output time", i.e., the time interval at which observables are computed and saved?

C. Now inspect the operator file. The part denoted `HAMILTONIAN-SECTION` contains the Hamiltonian operator, with the kinetic energy and potential operators given as in Eqs. (2.2) and (2.1), respectively. Each term of the Hamiltonian is given in a single line of the file.

The first terms in the Hamiltonian definition refer to the kinetic energy operator. In particular, the notation `KE` denotes the operator $-\frac{1}{2}\partial^2/\partial q^2$ in the `MCTDH` input; therefore, according to Eq. (2.2), the numerical coefficients in each of the terms containing the `KE` operator correspond to the mode frequencies in atomic units.

Next, the terms associated to the diabatic potential follows. Since the potentials differ for the two diabatic states each term is associated to an operator denoted as `S`α`&`β and associated with the "electronic" degree of freedom. In particular, `S1&1` denotes a term for the potential of the diabatic state 1 (B_{2g} in this case), `S2&2` is a terms for the potential of the state 2 (here B_{2u}), and `S1&2` (or, equivalently, `S2&1` is a term for the diabatic coupling. In Dirac notation, the expressions `S1&1`, `S2&2` and `S1&2` denote the Hermitian operators $|1\rangle\langle 1|$, $|2\rangle\langle 2|$ and $(|1\rangle\langle 2| + |2\rangle\langle 1|)$, respectively.

Question 5. Focus on the *zeroth order* purely electronic terms (i.e., those without any `q` or `q^2` operator). These terms coincide with the electronic Hamiltonian for the normal modes in their ground state equilibrium ($q_\kappa = 0$). Is there any diabatic coupling between the B_{2g} and B_{3g} states? Why?

Question 6. Focus on the linear terms associated with the operator `q`. Which mode numbers have linear functions for the diagonal potentials and the off-diagonal couplings? What is the symmetry of these modes?

D. After inspecting the input it is time to launch the quantum dynamical calculation. To this end, move to the folder where the files `C4H4_B2u.inp` and `C4H4.op` are located and launch the calculation as follows.

```
$ mctdh86 -w -mnd C4H4_B2u &
```

To see the meaning of the options `-w` and `-mnd` just type `mctdh86 -h` in the shell. The ampersand (`&`) runs the command in the background, allowing the shell to accept new commands immediately.

The `mctdh86` code has now created the folder `C4H4_B2u` inside which the calculation is running. It will take 20–30 minutes to complete, but, since it involves the propagation of the molecular wavefunction in steps, the results will become gradually available so they can be immediately inspected. Move into the folder `C4H4_B2u` and open the file `output` with an editor. For each output time (specified by the parameter `tout` in the input) some information are written.

Question 7. In the file `output`, what are the quantities indicated as `Norm`, `E-tot`, `state = 1 pop.`, `state = 2 pop.`, `<q>` and `<dq>`? What is their formal definition? Which of these quantities should be conserved over time if the propagation were performed with infinite integration accuracy and numerical precision?

E. During the wave packet propagation, the `MCDTH` program updates the file `auto`, which

contains the autocorrelation function S(t), defined as

$$\begin{aligned} S(t) &= \langle \Psi, 0 | \Psi, t \rangle \\ &= \left\langle \Psi, 0 \left| e^{-\frac{i}{\hbar}\hat{H}t} \right| \Psi, 0 \right\rangle , \end{aligned} \tag{2.5}$$

where $\hat{H}$ is the full vibrational-electronic (i.e., molecular) Hamiltonian. The autocorrelation function can be plotted using the script `plauto` (check `plauto -h` for all the options of the script). Alternatively, une case use gnuplot. For example the following commands will plot the absolute value $|S(t)|$ in the the range 0–200 fs.

```
$ gnuplot

gnuplot> plot [0:200] 'auto' using 1:4 with lines
# or, more compactly, "pl [0:200] 'auto' u 1:4 w l"
```

Question 8. Plot the real and imaginary parts of the autocorrelation function, as well as the absolute value. How do they look like? Does the function $|S(t)|$ display regular recurrences or is it rather noisy? Why? Why is its maximum well below 1 at all times (except for $t = 0$)?

The autocorrelation function is a superposition of oscillations whose frequencies correspond to energy eigenvalues of the coupled vibrational-electronic (non-Born-Oppenheimer) system, divided by $\hbar$. Therefore, it can be used to computed the vibronically resolved photoelectron spectrum $\sigma(\omega)$ (for the ionisation to the B_{2u} state) via Fourier transform,

$$\sigma(\omega) \sim \omega \mathrm{Re} \int_0^\infty S(t) e^{i\omega t} dt \, . \tag{2.6}$$

Question 9. Show that Eq. (2.6 corresponds to a sum of peaks in correspondence of the eigenvalues of the molecular Hamiltonian $\hat{H}$. What is the advantage of the time-dependent formalism as compared to the use of such sum-over-states expression?

In practice, Eq. (2.6) cannot be used because it would involve an integration up to infinity, while the autocorrelation function is available just up to a certain finite time T. Truncating this integral gives rise to spurious peaks in the Fourier transform, that can be eliminated by multiplying the autocorrelation function with a function $g(t)$ of the form

$$g(t) = \cos^n \left(\frac{\pi t}{2T} \right) \, . \tag{2.7}$$

In addition, before Fourier transform, $S(t)$ is also typically multiplied by a damping function – in this case a Gaussian – to replicate the experimental line broadening,

$$h(t) = \exp \left[- \left(\frac{t}{\tau} \right)^2 \right] . \tag{2.8}$$

In summary, a realistic spectrum, comparable to the experimental one, is evaluated from the formula

$$\sigma(\omega) \sim \omega \mathrm{Re} \int_0^T S(t) g(t) h(t) \mathrm{e}^{\mathrm{i}\omega t} \mathrm{d}t \,. \tag{2.9}$$

Using the `MCTDH` package the "spectrum-over-frequency" $\sigma(\omega)/\omega$ can be plotted or saved on a file using the scripts `plspec` or `autospec`, respectively. Note that, in this model, the Hamiltonian was shifted by a constant of $\approx 8\,\mathrm{eV}$, therefore the energy range is very different from the actual ionisation energy.

```
$ # Plot the spectrum using the g(t) function with n =2
$ # and the h(t) function with tau = 40 fs
$ # in the energy range 1.5 eV - 2.5 eV

$ plspec -g2   1.5 2.5 ev  40 2


$ # Save the spectrum to the file spectrum.pl

$ autospec86  1.5 2.5 ev 40 2
```

Question 10. Using the script `plspec`, adjust the frequency range so that the whole spectrum fits well in the plot, then save the final computed spectrum using `autospec`. How does the spectrum change when the parameter τ varies? Why?

F. The population of the electronic states is reported in the `output` file, but can be easily extracted using the script `rdcheck86` and plotted using `gnuplot` as follows

```
$ rdcheck86  1 0
# The diabatic populations are now reported in the file chk.pl

$ gnuplot
gnuplot> pl 'chk.pl' u 1:2 w l t 'B_{2g}', \
>   '' u 1:3 w l t 'B_{2u}'
```

Question 11. How does the electronic population dynamics look like? Describe how rapidly the population of the B_{2g} state builds up, whether the population transfer is oscillatory or monotonic, and how rapidly do the oscillations vanish.

Question 12. Is there a dominant period in the oscillations of the population? From the population plot, evaluate this period graphically, and calculate its frequency. Which modes have the closest frequency, therefore are most involved in promoting the population transfer? Remember that the mode frequencies are simply the coefficients of the "`KE`" terms in the operator file.
The coefficients of the linear terms for the initially populated state represent instead the initial "force" felt by the wave packet. Is the largest force along the same mode(s) the modulate the electronic transition?

G. The dynamical simulation performed with the `MCTDH` fully accounts for all possible quantum effects in the nuclear motion. As a next step, such quantum mechanical results are compared with those obtained via quantum-classical methods, based on the propagation of multiple trajectories. The first method to be used is the **fewest-switches surface hopping** (FSSH) approach.

FSSH simulations can be performed using the package `pysurf`. It is convenient to move to a new shell – keeping the shell of the MCTDH calculation open – and "activate" the PySurf Python environment using the `activate_pysurf` script. The successful activation is signaled by the appearance of the `(pysurf)` flag:

```
$ activate_pysurf
(pysurf) $
```

Once the environment is activated, move to the folder `exercise_1/fssh` and inspect the potential energy surface file `C4H4.pes`, which has same structure of the operator file for the `MCTDH` code. To generate the input file launch the script `Setup_Dynamics.py`, which is located in the folder `praktikum/scripts`:

```
(pysurf) $ python ../../scripts/Setup_Dynamics.py
```

The script will start asking a number of questions to set up the input file. Choose 100 trajectories, an integration time step of 0.1 fs, and choose the remaining parameters to be consistent with the previous MCTDH simulation (i.e., 200 fs of propagation time, output every 1 fs, etc.). Choose "Wigner" as a sampler and "spp.inp" when asked for surface point provider. The "number of states" is the number of electronic states and "active states" is a list of the states included in the simulation, counted starting from 0 (i.e., the answer is `0 1`). When asked for the input source for the normal modes choose `model`, then `SumOfProductPES`; the file for the parameters is `C4H4.pes`. Furthermore, since the potential is precomputed (i.e., not calculated "on-the-fly" by an electronic structure code) between `ab-initio` and `model` choose always `model`.

If everything goes smoothly, the script should create the input file `dyna.inp`, the "surface-point-provider" file `spp.inp` that contains instruction on how the potentials should be calculated, as well as 100 different folder, that will contain the information for the different trajectories. Each folder contains a `run.inp` file that includes the initial condition for each trajectory.

Question 13. Inspect the files `dyna.inp` and `spp.inp` created by the setup script. Each of them contains a parameter called `random_state` or `random_seed`, which is used to initialise a random number generator. Why is this potentially important?

At this stage, we are ready to launch the FSSH trajectories. This is done using the script `Launch_Dynamics.py`:

```
(pysurf) $ python ../../scripts/Launch_Dynamics.py \
 --i_traj 0 --f_traj 99 --method fssh
```

Here, the arguments `i_traj` and `f_traj` signal the first and last folder for which the trajectory is computed, and `method` selects a specific dynamical method (in this case FSSH).

Question 14. How many integration steps are performed for each trajectory? Which quantities, related to the electronic states and potential energy surfaces, need to be computed at each step to advance the trajectory?

Question 15. As you might have noticed, running all the trajectories takes several minutes, even if a precomputed surface is used. How could one easily decrease the total simulation wall time?

H. Once the trajectories are computed one can evaluate the electronic populations by taking an average over the trajectories, and compare them with the quantum dynamical result. This is performed using the script `Statistics_TSH.py`. However, one has to consider that the populations obtained by quantum refer to the *diabatic* states, whereas the surface hopping simulation is performed in the *adiabatic* representation. Adiabatic populations are obtained directly from the coefficients of Eq. (1.5) as

$$P_\alpha(t) = \frac{1}{N_{\text{traj}}} \sum_{j=1}^{N_{\text{traj}}} \left| C_\alpha^{(j)}(t) \right|^2 , \tag{2.10}$$

where $C_\alpha^{(j)}(t)$ is the coefficient for the adiabatic state α for the j-th trajectory at time t. The diabatic populations are obtained by an identical formula, with $C_\alpha^{(j)}$ replaced by the diabatic coefficient $\widetilde{C}_\alpha^{(j)}$, given as

$$\widetilde{C}_\alpha^{(j)}(t) = \sum_\beta U_{\alpha\beta}\left(\mathbf{q}^{(j)}(t)\right) C_\beta^{(j)}(t) , \tag{2.11}$$

where $\mathbf{q}^{(j)}(t)$ is the vector of the coordinates for the trajectory j at time t and $U_{\alpha\beta}(\mathbf{q})$ is the geometry-dependent *adiabatic-to-diabatic* transformation matrix .

Question 16. From $\mathbf{q}^{(j)}(t)$ one can compute the diabatic electronic Hamiltonian matrix $W_{\alpha\beta}$ from Eq. (2.1). How is the matrix $U_{\alpha\beta}$, needed in Eq. (2.11), then computed?

All these steps are performed by the script `Statistics_TSH.py` by invoking the option `--diabatic`:

```
(pysurf) $ python ../../scripts/Statistics_TSH.py \
 --i_traj 0 --f_traj 99 --diabatic
```

The script generates a file called `population.out` that contains the populations of the different electronic states, computed in different ways (see the first line of the file for more information). In particular, using `gnuplot`, we can compare the adiabatic and diabatic populations for the state that was initially unpopulated:

```
(pysurf) $ gnuplot

gnuplot> set key b   # place the legend at the bottom
gnuplot> set xlabel 'time␣[fs]'
gnuplot> set ylabel 'population'
gnuplot> pl 'population.out' u 1:4 w l t 'S_1␣(adiabatic)', \
      '' u 1:8 w l t 'B_{2g}␣(diabatic)'
```

Question 17. How do the populations look like? In the diabatic representation, does one state get predominantly populated, or the two states get similar populations? How does this change in the adiabatic representation? Compared to the diabatic case, does the adiabatic population oscillate considerably after 20–30 fs? Does this imply that the Born-Oppenheimer approximation holds at these times?

I. Finally, we perfom an Ehrenfest calculation, and then compare the three methods (quantum dynamics, fewest-switches surface hopping, Eherenfest). This is easier, because we can reuse the same input files (`dyna.inp`) and `spp.inp` that we used for the FSSH simulation. To this end, move into the folder `ehrenfest` and copy the twp files from the folder `fssh`.

At this stage, one can generate the trajectory folders and launch the Ehrenfest calculations using the same commands as before:

```
(pysurf) $ python ../../scripts/Setup_Dynamics.py

(pysurf) $ python ../../scripts/Launch_Dynamics.py \
 --i_traj 0 --f_traj 99 --method ehrenfest
```

Note that this time no questions are asked by the `Setup_Dynamics.py` script.

Question 18. While the trajectories are running, inspect the initial conditions (positions and momenta) for a certain trajectory number by opening the file `traj_000000xx/run.inp` in both the folders `fssh` and `ehrenfest`. Check that the initial conditions are the same, since we used exactly the same input. Why is this important?

After all trajectories have been computed, the electronic populations can be evaluated using the script `Statistics_Mapping.py`. In this case no additional flags are needed, since the script calculates automatically the diabatic populations and prints the results in the file `population_diab.out`.

```
(pysurf) $ python ../../scripts/Statistics_Mapping.py \
 --i_traj 0 --f_traj 99
```

Question 19. Compare the diabatic populations obtained with the FSSH and the Ehrenfest approaches against the quantum dynamical reference result. Which quantum-classical methods has a better agreement? Summarise the qualitative and quantitative differences between the various approaches.

Tutorial 2. Nonadiabatic dynamics of pyrazine

The lowest energy band of the electronic absorption spectrum of pyrazine involves two electronic states that, at D_{2h} geometries, have B_{2u} and B_{3u} symmetry. Different electronic structure approaches predict different energetic spacing between the two states; in some studies a third state, of A_u symmetry, has also been proposed to contribute to the band.[1][2] Therefore, despite its apparent simplicity, the vibrational-electronic structure of pyrazine has been a challenge for computational photochemistry for decades.

In this exercise the vibronic absorption spectrum of pyrazine and the underlying molecular photodynamics are investigated. The main goals are:

1. Understand the effect of the vibronic coupling on the spectral features;

2. Predict how the photodynamics changes upon varying the energy gap between the B_{2u} and B_{3u} states, and understand whether quantum-classical simulations reproduce the correct trend obtained by a full quantum mechanical calculation.

To this end, nonadiabatic molecular dynamics simulations are performed based on a well established four-modes linear vibronic coupling model.[3] The model is set up in dimensionless normal modes, so that the kinetic energy operator and the initial state have the same form as for butatriene – see Eqs. (2.1) and (2.3).

The excited state diabatic potentials have the same frequencies and normal modes of the ground state and are simply linearly displaced. Denoting with "1" and "2" the states B_{3u} and B_{2u}, respectively, the diabatic potentials and couplings have the form

$$W_{11}(\mathbf{q}) = -\Delta + \sum_{j=1}^{4} \kappa_j^{(1)} q_j + \sum_{j=1}^{4} \frac{\hbar\omega_j}{2} q_j^2 \,, \tag{3.1a}$$

$$W_{22}(\mathbf{q}) = \Delta + \sum_{j=1}^{4} \kappa_j^{(2)} q_j + \sum_{j=1}^{4} \frac{\hbar\omega_j}{2} q_j^2 \,, \tag{3.1b}$$

[1] M. Sala, B. Lasorne, F. Gatti and S. Guérin, *Phys. Chem. Chem. Phys.* **16**, 15957 (2014)
[2] P. Wójcik, H. Reisler, J. F. Stanton and A. I. Krylov, *J. Chem. Phys.* **163**, 054110 (2025)
[3] G. A. Worth, H.-D. Meyer and L. S. Cederbaum, *J. Chem. Phys.* **109**, 3518 (1998)

$$W_{12}(\mathbf{q}) = \sum_{j=1}^{4} \lambda_j q_j \,. \tag{3.1c}$$

Question 1. What is the symmetry of the diabatic coupling function $W_{12}(\mathbf{q})$?

It is assumed that the initial state is prepared by an instantaneous excitation (δ in time) from the ground electronic state to the B_{2u} state. The ground state potential is not included in the simulation, but it's assumed to be simply given as $W_{gs}(\mathbf{q}) = E_{gs} + \frac{1}{2}\sum_{j=1}^{4} \hbar\omega_j q_j^2$.

A. We start from the quantum dynamical simulation of the $B_{2u} \longrightarrow B_{3u}$ relaxation. Move to the folder `exercise_2/quantum_dynamics` where the input file for the `MCTDH` program (`pyr4.inp`) and the operator file (`pyr4.op`) are located. Inspect the operator file to understand its structure; note that the parameters are given in a separate section.

Question 2. What are the frequencies of the four modes, in cm^{-1}? For which modes the parameters $\kappa_j^{(1)}$, $\kappa_j^{(2)}$ and λ_j are different from zero? What is the value of Δ?

Question 3. From Eqs. (3.1a) and (3.1b) and using the parameters of the operator file, compute the values of the coordinates q_j corresponding to the minima of the diabatic potentials W_{11} and W_{22}. Which mode is the most displaced with respect to the ground state minimum, located at $q_j = 0$?

In this exercise it is useful to plot the diabatic potentials along the most displaced mode. At the step ?? this will be repeate for different values of Δ, and the plots are typical done by setting all other modes to their value at the ground state minimum. Feel free to use your favourite plotting program to plot the potentials. Using `gnuplot`, a nice figure of the potentials along the q_{6a} mode can be obtained with the following commands[4] (search online the meaning of the `gnuplot` options `lw`, `lc` and `dt`):

[4]The energy conversion factor, from hartrees to eV, is taken from the NIST website.

```
gnuplot> set term qt font 'Arial,␣20'
Terminal type is now 'qt'
Options are '0␣font␣"Arial,20"'

gnuplot> au2eV = 27.211386

gnuplot> w6a    = 0.0740 / au2eV
gnuplot> k6a1   =-0.0964 / au2eV
gnuplot> k6a2   = 0.1194 / au2eV
gnuplot> delta  = 0.46165 / au2eV

gnuplot> W_11(x) = - delta + k6a1*x + 0.5*w6a * x**2
gnuplot> W_22(x) = + delta + k6a2*x + 0.5*w6a * x**2

gnuplot> set xlabel 'q_{6a}'
gnuplot> set ylabel 'energy␣[eV]'
gnuplot> set key b l   # legend at the bottom-left

gnuplot> pl [-6:6] [-1:1.5] (W_11(x) * au2eV) \
         w l lw 3 lc '#ff0000' t 'B_{3u}', \
         (W_22(x) * au2eV) w l lw 3 lc '#008000' t 'B_{2u}', \
         (delta * au2eV) w l lw 2 lc '#808080' dt 2 t ''
```

B. The next step is to launch the wave packet simulation, and compute the resulting diabatic population dynamics and the linear absorption spectrum. To perform these tasks, follow the indications of the steps **D**, **E** and **F** of tutorial 1.

Question 4. What is the total propagation time requested in the input file? What is the output time? Describe the structure of the MCTDH wavefunction in terms of number of single-particle functions for each mode and each diabatic state, as well as the number of grid points.

Question 5. What are the recurrence periods for the frequencies of the displaced modes? Plot the absolute value of the autocorrelation function (4th column in the file `auto`). Do the vibrational periods match the recurrence times present in the autocorrelation function?

The spectrum is evaluated according to Eq. (2.9) using the script `autospec86`. Here we use

the purely exponential decay

$$h(t) = \exp\left(-\frac{t}{\tau}\right) \tag{3.2}$$

to damp the autocorrelation function.

Question 6. What is the line shape $\sigma_h(\omega)$ associated to the exponential decay of Eq. (3.2)? To find it, evaluate the real part of the (half-)Fourier transform of the decay function, i.e.

$$\sigma_h(\omega) = \int_0^\infty h(t) e^{i\omega t} dt \,. \tag{3.3}$$

Question 7. Analyse the computed spectrum. How many electronic bands do you see? Is this consistent with the fact that, in the simulation, it is assumed that the only allowed transition is to the diabatic B_{2u} state?

Question 8. How does the diabatic population curve look like? What is the timescale for the $B_{2u} \longrightarrow B_{3u}$ transfer? Are there recurrences? What is the maximum amount of transfer?

C. As a next step, we check how the absorption spectrum changes when the diabatic coupling is neglected. To this end, move back to the folder `exercise_2/quantum_dynamics`, make a copy of the input and operator files:

```
$ cp pyr4.inp pyr4_no_coupling.inp
$ cp pyr4.op pyr4_no_coupling.op
```

Now, edit the operator file `pyr4_no_coupling.op` to remove the `S1&2` coupling. It is sufficient to set the parameter `lambda` to zero.

In a similar way, edit the input file `pyr4_no_coupling.inp`: change the `name` keyword to `pyr4_no_coupling` and `opname` to `pyr4_no_coupling`. Now re-launch the MCTDH calculation for this new input and compute the spectrum.[5]

Question 9. How does the absorption spectrum change when the diabatic coupling is neglected?

D. The `MCTDH` program allows the visualisation of the vibrational densities of the wave packet, for each diabatic component, along the various modes. It is interesting to inspect

[5]Note that, in this case, it is sufficient to use a single SPF for each degree of freedom, because the dynamics occurs on a single separable potential energy surface and the initial wavefunction is in a product form.

the density for the case where the diabatic coupling is nonzero. In this case, the vibrational wavefunctions associated to the different diabatic states are rather different.

Question 10. Denoting as $\chi^{(\alpha)}(q_1, ..., q_4, t)$ the vibrational wavefunction associated with the diabatic state $|\alpha\rangle$, provide a formula for the vibrational density $\varrho(q_1, t)$ for the first mode (denoted "10a" in the input files).

The time evolution of the vibrational densities for the two diabatic states can be plotted using the `showd1d` script (check `showd1d86 -h` to display information about the various options):

```
# Vibrational density for the mode 1 and the electronic state 2
$ showd1d86 -sm -M -a -y 0.7 f1 s2

# Vibrational density for the mode 1 and the electronic state 1
# printed to an ascii grid file
$ showd1d86 -T f1 s1
```

Question 11. Describe the shape and evolution of the vibrational densities for the mode 1. How are the densities normalised? Is there a node conserved during the time evolution? What is the reason for this node?

Hint: To understand the node structure, think about the total symmetry of the molecular state, which must be conserved during the dynamics because the Hamiltonian is totally symmetric. Alternatively, you can check the short time dynamics analytically,

$$|\Psi, \delta t\rangle \approx |\Psi, 0\rangle - \frac{\mathrm{i}\delta t}{\hbar}\hat{H}|\Psi, 0\rangle \,.$$

Does this "short time" wavefunction have a node?

The `-T` option of the script `showd1d86` produces a file (typically called `den1d_fx_sx`) that contains the density in a coordinate-time grid. This can be useful to plot the density with a separate program. For example, the following commands produce a 2D map of the density using `gnuplot`:

```
gnuplot> set view map
gnuplot> unset surface
gnuplot> set pm3d

gnuplot> set xlabel 'time␣[fs]'
gnuplot> set ylabel 'q_1'

gnuplot> set pm3d interpolate 4,4
gnuplot> set palette defined (0 'white', 0.5 'blue', 1 'black')
gnuplot> set cbrange [0:0.6]

gnuplot> splot 'den1d_f1_s1' u 2:1:3
```

E. The linear vibronic coupling model used for this tutorial was constructed from electronic structure data, such as electronic excitation energies as a function of the molecular geometry. Depending on the level of theory, the electronic energy gap between the B_{2u} and B_{3u} states can be off by $0.1 - 0.5$ eV. Therefore it makes sense to study how the predicted population dynamics would change for different values of the energy gap.

To this end, we repeat the simulation of the population dynamics by changing the parameter Δ of Eqs. (3.1a) and (3.1b) to the following values: 0.0 eV, 0.2 eV, 0.4 eV, 0.6 eV and 0.8 eV.

Question 12. Repeat the quantum dynamical calculation by changing the values of Δ. This is conveniently done by preparing new input and operator files (be sure that each calculation has its own directory!). For each calculation, use the command `rdcheck86 1 0` to generate a `chk.pl` file, that contains the diabatic populations, and plot them for the different values of Δ in the same plot.
How does the population dynamics change as a function of Δ? For which value(s) is the population transfer maximum in the first 50, or 200 fs? Which value corresponds to the slowest transfer (where, e.g., it takes the longest time to achieve at least 20% of transfer)?

F. At this stage it is interesting to compare the whether quantum-classical surface hopping simulations reproduce the same trends obtained by the full quantum dynamical simulations. To this end, move to the folder `exercise_2/fssh_0.4` which contains an operator file (`pyr4.inp`) for a calculation to be performed using `pysurf`.

Inspect the file `pyr4.inp`. It has a different structure than the file used in the Tutorial 1, and contains the parameters of the linear vibronic coupling model in atomic units (hartrees).

In particular, the variable `W0` contains the zeroth order diabatic energies, i.e. the parameter Δ.

To launch the FSSH calculation, follow the same steps of the point **G** of Tutorial 1. When asked for the model, choose `LVC` instead of `SumOfProductPES`. Use 200 fs of propagation time with a time step of 0.1 fs and an output time of 1 fs. Create the trajectories folders with the `SetupDynamics.py` script, launch with `LaunchDynamics.py` and analyse with `Statistics_TSH.py` (don't forget the option `-diabatic` to evaluate populations comparable with those computed via quantum dynamics).

Question 13. In different folders (`fssh_0.0`, `fssh_0.2`, etc.) perform different FSSH simulations for the values $\Delta = 0.0\,\text{eV}, 0.2\,\text{eV}, 0.4\,\text{eV}, 0.6\,\text{eV}, 0.8\,\text{eV}$ by editing the `pyr4.inp` potential file (be aware that the parameter Δ should be given in atomic units). Analyse the population dynamics, plotting the population of the initially "empty" states in the same plot for different values of Δ. Do the FSSH calculation reproduce the same trend obtained by quantum dynamics? What are the main differences?

G. To complete the comparison between quantum and quantum-classical simulations, one can inspect the vibrational densities. For the FSSH calculations, the densities can be computed using the script `Statistics_TSH_density.py`. The provided script works automatically in the diabatic representation and takes the mode and state number as input.

Consider that, according to the Python conventions, the numbering starts from 0; therefore, for example, the following command computes the density along the mode 1 (Q_{10a}) for the B_{3u} state on a grid of 57 points from -7 to 7:

```
(pysurf) $ cd fssh_0.4
(pysurf) $ python ../../scripts/Statistics_TSH_density.py \
   --i_traj 0 --f_traj 99 --mode 0 --state 0 \
   --grid "'-7.0 7.0 57'" --sigma 0.5
```

This command produces the file `density_0_0.py` which contains 3D grid data and can be plotted using gnuplot in the same way describe at the end of point **D**. The parameter `sigma` is used to broaden the classical distribution to mimic the quantum mechanical one – the larger the value the larger the broadening.

Question 14. How does the quantum-classical density compare with the full quantum dynamical one? Does it exhibit the same recurrences? Does it also display a nodal line?

Tutorial 3. Na + Cl collision

This tutorial explores the nonadiabatic dynamics of alkyl halides—particularly sodium chloride—which have served as prototypical examples for studying the coupling between nuclear and electronic motion. The study of these systems was instrumental in the development of the earliest femtosecond spectroscopic techniques in the late 1980s and led Ahmed Zewail to win the Nobel Prize in Chemistry in 1999.[1]

The goal of this tutorial is to investigate the probability of inducing an electronic transition between the two lowest electronic states of NaCl during a rectilinear collision between neutral Na and Cl atoms. This probability is evaluated as a function of the collision speed through three steps.

1. First, the dependence of the electronic transition probability on the collision speed is calculated analytically in the quantum–classical limit using the so-called Landau–Zener model, assuming weak electronic coupling between the two diabatic states.

2. In a second step, quantum dynamical simulations are used to predict the transition probability, and the results are compared with the analytical quantum–classical predictions.

3. Finally, the quantum mechanical results are compared with those obtained from Landau–Zener surface-hopping simulations.

As mentioned above, the molecular Hamiltonian includes two diabatic states, associated with *ionic* and *covalent* electronic configurations, respectively. At short Na–Cl distances, the lowest adiabatic state is predominantly ionic in character. Upon dissociation (i.e., at large Na–Cl distances), the lowest adiabatic state coincides with the covalent diabatic state. The calculations are based on diabatic potential energy curves constructed by Giese and York using electronic structure calculations at the multireference configuration interaction level of theory.[2]

A. As a first step, it is useful to become familiar with the potential energy curves for the ionic and covalent states. These are tabulated (in hartrees) on a grid of Na–Cl distances

[1] See A. Zewail's Nobel lecture: https://www.nobelprize.org/uploads/2018/06/zewail-lecture.pdf

[2] T. J. Giese and D. M. York, *J. Chem. Phys.* **120**, 7939 (2004)

(given in bohr) in the files `w_11.dat`, `w_22.dat`, and `w_12.dat`. Using `gnuplot`, the diabatic potentials for the ionic and covalent states can be plotted as follows:

Question 1. Plot the diabatic potentials in the files `w_11.dat` and `w_22.dat`. Which one corresponds to the ionic state, and which corresponds to the covalent state? How is the zero of the energy scale defined? Which power n in the expression $1/R^n$ describes the shape of the ionic potential at large Na–Cl distances R?

For a molecule with two electronic states, the electronic Hamiltonian (i.e. the full molecular Hamiltonian minus the nuclear kinetic energy) can be represented by a 2×2 diabatic potential matrix

$$\hat{H}_{\text{el}} = \begin{pmatrix} W_{11}(R) & W_{12}(R) \\ W_{12}(R) & W_{22}(R) \end{pmatrix} , \tag{4.1}$$

and the potential energy curves of the adiabatic states (i.e. the eigenstates of the electronic Hamiltonian) are given by the eigenvalues of the matrix in Eq. (4.1).

Question 2. Derive the expressions for the adiabatic potentials $V_1(R)$ and $V_2(R)$ as the eigenvalues of the symmetric 2×2 electronic Hamiltonian matrix and plot them. To this end, you can use the file `w.dat`, which contains the values of $W_{11}(R)$, $W_{22}(R)$, and $W_{12}(R)$ in columns 2, 3, and 4. In `gnuplot`, it is possible to perform operations on data in different columns using the identifier `$`. For example, the average potential $(W_{11}(R) + W_{22}(R))/2$ can be plotted with the command:

```
gnuplot> pl 'w.dat' u 1:(0.5 * ($2 + $3)) w l
```

Do the adiabatic potentials exhibit an intersection point, or do they instead display an "avoided crossing"?

B. Near the ionic/covalent crossing, the diagonal diabatic potentials can be approximated as straight lines with slopes κ_1 and κ_2, and the diabatic coupling can be taken to be a constant λ. Within this model, consider a Na–Cl collision initiated at infinite separation, in which the two atoms approach each other with a relative velocity v and reach the diabatic crossing point at time $t = 0$. These approximations define the so-called **Landau–Zener model**.

Question 3. Write the expressions for the diabatic potentials $W_{11}(R)$ and $W_{22}(R)$ in the Landau–Zener model. Assume that they are straight lines with slopes κ_1 and κ_2, crossing at the distance R_0 with energy E_0.
There is no need to introduce numerical values for the parameters; treat all of them as variables.

In a quantum–classical description of the collision, the electronic state is a time-dependent superposition of the ionic and covalent configurations, denoted by the kets $|1\rangle$ and $|2\rangle$,

$$|\psi_{\text{el}}, t\rangle = C_1(t)|1\rangle + C_2(t)|2\rangle \,, \tag{4.2}$$

and the time-dependent coefficients satisfy the coupled time-dependent Schrödinger equations

$$\begin{pmatrix} \frac{\mathrm{d}C_1}{\mathrm{d}t} \\ \frac{\mathrm{d}C_2}{\mathrm{d}t} \end{pmatrix} = -\frac{\mathrm{i}}{\hbar} \begin{pmatrix} W_{11}\left(R(t)\right) & W_{12}\left(R(t)\right) \\ W_{12}\left(R(t)\right) & W_{22}\left(R(t)\right) \end{pmatrix} \begin{pmatrix} C_1(t) \\ C_2(t) \end{pmatrix} . \tag{4.3}$$

Question 4. Write the explicit equations of motion for $C_1(t)$ and $C_2(t)$ by performing the matrix-vector multiplication in Eq. (4.3). Use the linear functions derived in **Question 3** for the diagonal potentials and the constant λ for the coupling. Finally, take a rectilinear trajectory for $R(t)$ with constant velocity v that reaches the crossing R_0 at time $t = 0$.The resulting equations of motion should have the following form:

$$\frac{\mathrm{d}C_1}{\mathrm{d}t} = f_1(t)C_1(t) - \frac{\mathrm{i}\lambda}{\hbar}\lambda C_2(t) \,, \tag{4.4a}$$

$$\frac{\mathrm{d}C_2}{\mathrm{d}t} = f_2(t)C_2(t) - \frac{\mathrm{i}\lambda}{\hbar}C_1(t) \,. \tag{4.4b}$$

What are the functions $f_1(t)$ and $f_2(t)$?

Suppose that the system is initially (at "minus infinite" time) prepared in the diabatic state $|2\rangle$. Then we can set $C_1(-\infty) = 0$ and $|C_2(-\infty)| = 1$. The final population transferred to the diabatic state $|1\rangle$ (at "plus infinite" time) can be obtained by solving Eqs. (4.4a) and (4.4b) with these initial conditions. This can be done analytically, but the solution of the two differential equations is mathematically rather advanced. What is much easier to do is solving Eqs. (4.4a) and (4.4b) in the limit of weak coupling, i.e., for a small value of λ, using pertubation theory.

Question 5. To obtain a perturbative expression for the population of the state $|1\rangle$ after passage through the diabatic crossing, follow these steps:

1. Introduce the new variables

$$\widetilde{C}_1(t) = \mathrm{e}^{-F_1(t)} C_1(t) \,, \tag{4.5a}$$

$$\widetilde{C}_2(t) = \mathrm{e}^{-F_2(t)} C_2(t) \,, \tag{4.5b}$$

   where $F_1(t)$ and $F_2(t)$ are primitives of $f_1(t)$ and $f_2(t)$, i.e., they are defined as the indefinite integrals

$$F_\alpha(t) = \int f_\alpha(t) \mathrm{d}t \,. \tag{4.6}$$

   By calculating $F_1(t)$ and $F_2(t)$, obtain explicit expressions for $\widetilde{C}_1(t)$ and $\widetilde{C}_2(t)$. What are the absolute values $\left|\widetilde{C}_1(-\infty)\right|$ and $\left|\widetilde{C}_2(-\infty)\right|$?

2. Now derive an equation of motion for the variable $\widetilde{C}_1(t)$ by combining Eqs. (4.4a), (4.4b), (4.5a), and (4.5b), and by substituting the explicit expressions for the functions $f_1(t)$, $f_2(t)$, $F_1(t)$, and $F_2(t)$. If the derivation is correct, the final equation of motion will have the form

$$\frac{\mathrm{d}\widetilde{C}_1}{\mathrm{d}t} = a(t)\widetilde{C}_2(t) \,. \tag{4.7}$$

   What is $a(t)$?

3. Equation (4.7) can now be solved perturbatively. To this end, integrate both sides of the equation from $-\infty$ to $+\infty$, setting $\widetilde{C}_2(t)$ equal to its initial value on the right-hand side,

$$C_1(+\infty) = \int_{-\infty}^{+\infty} \mathrm{d}\tau \, a(\tau)\widetilde{C}_2(-\infty) \,. \tag{4.8}$$

   By evaluating the integral, derive a final expression for the population $|C_1(+\infty)|^2$ of the diabatic state $|1\rangle$.

   Hint: To evaluate the integral, it may be useful to rely on the following Gaussian integral:

$$\int_{-\infty}^{+\infty} \mathrm{e}^{-\alpha x^2} \mathrm{d}x = \sqrt{\frac{\pi}{\alpha}} \,, \tag{4.9}$$

   which—for the purposes of this exercise—can be considered valid also for complex values of α.

   How does the final population depend on the initial collision speed v?

C. The relation found in step **B** between the collision speed and the population transferred to the state $|1\rangle$—which is based on an idealised quantum–classical model—can be tested against full dynamical simulations. We begin with quantum dynamics. The folder `exercise_3/quantum_dynamics` contains the input and operator files for the `MCTDH` code, together with copies of the `.dat` files for the potentials.

Question 6. Inspect the operator file. Note that the diabatic potentials are read directly from the data files and interpolated onto the coordinate grid used for the calculation. The `PARAMETER-SECTION` also includes a mass m_R for the degree of freedom R, which the program uses to define the `KE` operator as $-\hbar^2/(2m_R)\,\partial^2/\partial R^2$. What does the value of this mass correspond to?

Question 7. Inspect the input file. Why do you think the propagation time is so long compared to the timescale of the photodynamics of the organic molecules studied in Tutorials 1 and 2?
Note that the operator section contains a *complex absorbing potential* (CAP). This is an imaginary potential term added to the Hamiltonian, and it has the form

$$V_{\mathrm{CAP}}(R) = \begin{cases} 0 & \text{for } R < R_{\mathrm{CAP}} \\ \mathrm{i}\eta\,(R - R_{\mathrm{CAP}})^n & \text{for } R \geq R_{\mathrm{CAP}} \end{cases} \tag{4.10}$$

(in this specific case, in atomic units, the parameters are $\eta = 0.3$, $R_{\mathrm{CAP}} = 28$, and $n = 2$). The CAP has the effect of damping the amplitude of the wavefunction as soon as it enters the region $R \geq R_{\mathrm{CAP}}$ (so the norm of the wavefunction decays in this region). Why is this important? What could happen if no CAP were present? Keep in mind that the coordinate grid used in the simulation is finite.

The `INIT_WF-SECTION` of the input file specifies an initial state located in the covalent diabatic state (i.e. the colliding atoms are neutral), and the nuclear wavefunction is taken to be a Gaussian with a width of $0.5\,a_0$, centred at $R = 24\,a_0$, and with a momentum of $P_0 = 4\,\hbar/a_0$ in the direction of the collision (hence the minus sign). The momentum is directly proportional to the collision velocity.

D. To test the relation found at the end of point **B**, it is necessary to vary the initial momentum and perform several quantum dynamical simulations. To this end, create five copies of the input file in which the initial momentum P_0 takes the values -4, -6, -8, -10, and -12. Note that the keyword `name` must be changed in each input file in order to run the simulations in different directories. Following the same procedures as in Tutorials 2 and 3, compute the population dynamics for each initial condition. Note that the `.dat` files containing the diabatic potentials must be copied into each directory.

Question 8. Plot the population of the ionic state (which is initially unoccupied) for the five values of P_0. Identify the timescales required to reach the avoided crossing, and focus on the first transfer event. How much population is transferred to the ionic state in each case? Plot the transferred population as a function of the initial momentum P_0 (or some suitable power P_0^k). Does it follow the same dependence predicted by the Landau–Zener model in step **B**?

E. Finally, the exact quantum mechanical populations are compared with quantum–classical results. To this end, we use a special variant of surface hopping that is based on the Landau–Zener model and is known as Landau–Zener surface hopping (LZSH).

To prepare the surface-hopping calculations, move to the folder `exercise_3/lzsh_4`. This folder contains the `NaCl.pes` file, which provides information about the potential energy surfaces, as well as the `.dat` files containing the potentials.

Prepare the `dyna.inp` and `spp.inp` files as usual using the script `SetupDynamics.py`. Choose 200 trajectories, a total propagation time of 2000 fs, an output time of 2 fs, and a time step of 1 fs. The number of states is 2, the list of states is `0 1`, and state 1 in the diabatic representation is initially populated. When asked about applying a shift, reply `yes`, because otherwise the initial density distribution would be centred at $R = 0$ and would have zero average momentum. The potential type is `SumOfProductsPES`, and the parameter file is `NaCl.pes`. When prompted, set `shift_q` to `24` and `shift_p` to `-4` to be consistent with the quantum dynamical simulations.

To set up simulations for different initial momenta, create additional folders and copy the files `dyna.inp`, `spp.inp`, and `w_*.dat` into each of them. In each folder, edit the file `dyna.inp` and set the appropriate value of the parameter `shift_p`, which determines the initial momentum.

Question 9. Launch the five LZSH simulations and evaluate the populations using the script `Statistics_LZ.py` (its usage is the same as that of the script `Statistics_TSH.py`; do not forget the option `--diabatic`). How do the resulting populations look? Do they reproduce the same trends as the classical simulations?

You will find that the quantum–classical simulations produce rather noisy results. This is because the population in the ionic state is of the order of 0.1–0.2, and the maximum statistical accuracy attainable with 200 trajectories is of comparable magnitude, $1/200 = 0.005$. This is a typical *rare-event sampling* problem. For a specific initial momentum, it may therefore be worthwhile to repeat the calculation with a larger number of trajectories. A good strategy is to launch 1000 trajectories in five batches of 200 each, running in parallel.